

BME 140 W24

Lab #6 Report

Lab: B1-2

ECG Signal Measurement

Brian Ngo

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## Abstract

In this laboratory experiment, we investigate the functionality of an ECG circuit designed in a previous exercise by evaluating it on ourselves. Through the placement of three electrodes on our bodies and their connection to the ECG, we explore the relationship between these electrodes. We discovered that data from Leads I and II can be leveraged to compute information for Lead III, effectively filtering noise around the PQRST wave and producing a clearer ECG waveform. By analyzing data from all three leads, we determine the mean electrical orientation, providing valuable insights into cardiac activity and the functioning of the heart.

## Introduction

The primary objective of this laboratory experiment is to understand bioelectrical signals and Electrocardiogram (ECG) analysis. We aim to comprehend the fundamental principles underlying bioelectric signals and the intricacies of ECG waveforms. By utilizing the electrocardiogram constructed in the previous lab session, we aim to capture and analyze variations in the bio potentials of the heart under different conditions, such as rest and muscle contraction. Electrodes are placed on the subject's wrists serve as sensors, detecting waveforms generated by the bio potentials within the body and forming the characteristic PQRST complex. Through analysis of the data collected from various leads, we will discover the mean electric orientation, providing insights into the polarization direction of the heart's electrical activity.

## Theoretical Background and Design

In the context of electrocardiography (ECG), the completed ECG circuit serves as a vital tool for amplifying bio-potentials within the body to facilitate analysis. The resulting ECG signal exhibits a periodic PQRST complex, which provides essential insights into cardiac activity and health status. Each segment of the PQRST wave corresponds to specific cardiac events: the P-wave signifies atrial depolarization and contraction, the QRS segment denotes ventricular depolarization and contraction coupled with atrial repolarization and relaxation, while the T wave represents ventricular repolarization and relaxation.

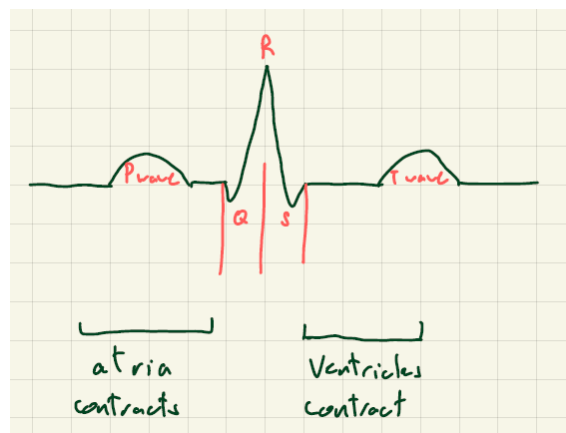


Figure 1: Basic PQRST Complex

Figure 2 illustrates the ECG circuit designed in previous sessions, demonstrating its capability to capture and amplify bio-potentials. These signals are acquired through electrodes placed on bipolar limb Leads I, II, and III, as depicted in Figure 3. Lead III, crucial for comprehensive ECG analysis, is calculated from Leads I and II using Kirchoff's Law:  $\text{Lead III} = \text{Lead I} - \text{Lead II}$ . This calculation enhances signal fidelity by filtering noise and refining the ECG waveform for precise interpretation.

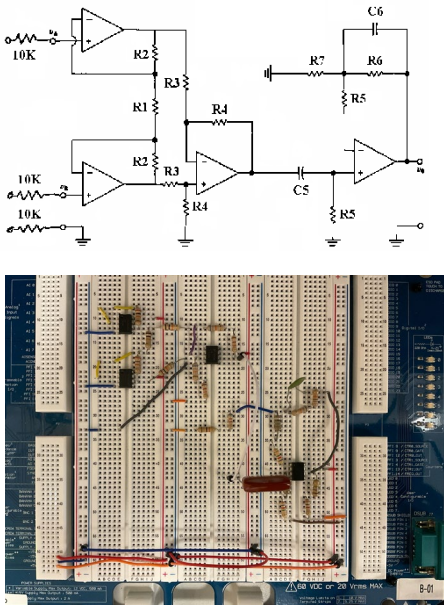


Figure 2: ECG Amplifier Schematic and Wiring

|       |       |       |       |         |        |         |      |         |
|-------|-------|-------|-------|---------|--------|---------|------|---------|
| R1    | R2    | R3    | R4    | R5      | R6     | R7      | C5   | C6      |
| 10 kΩ | 20 kΩ | 10 kΩ | 80 kΩ | 15.9 MΩ | 159 kΩ | 6.63 kΩ | 1 μF | 0.01 μF |

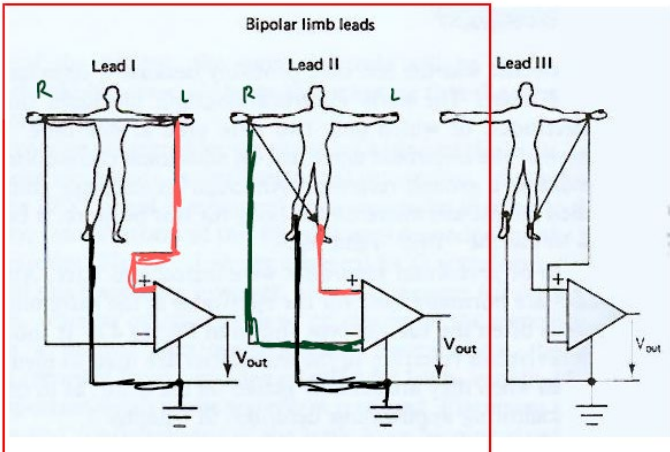


Figure 3: ECG Leads

The outputs from Leads I, II, and III enable determination of the mean electrical axis, elucidating the direction of depolarization and repolarization within the heart. By analyzing the maximum value of the QRS portion of the complex, clinicians and researchers can quantify the magnitude of depolarization and repolarization events, providing crucial diagnostic information.

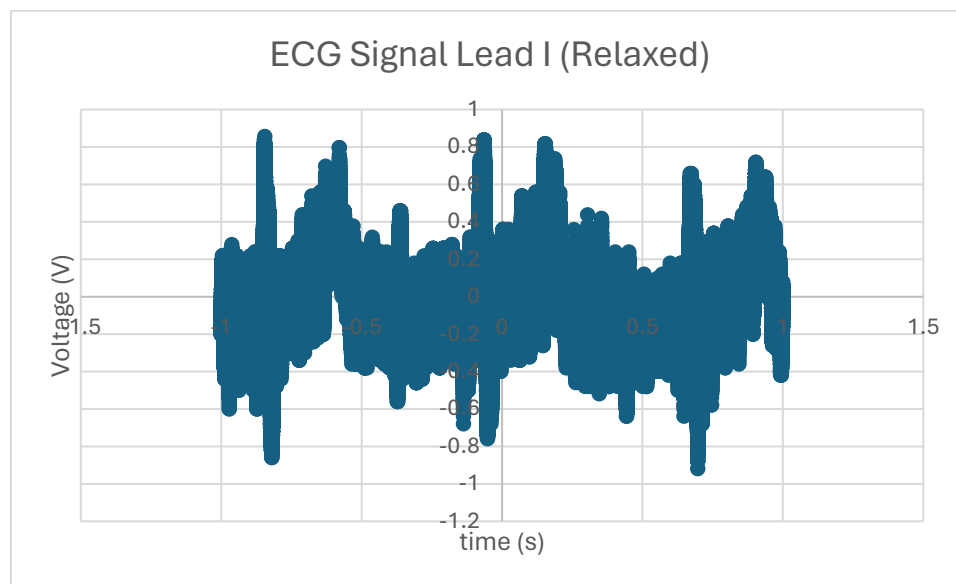
The theoretical foundation underpinning ECG signal analysis encompasses the principles of bioelectricity, cardiac physiology, and signal processing. By leveraging the insights gleaned from the PQRST complex and the calculated Leads, clinicians can accurately diagnose cardiac abnormalities, assess heart health, and formulate effective treatment strategies. Through the integration of theoretical concepts and practical circuit design, this laboratory experiment offers a comprehensive understanding of ECG signal acquisition and analysis, empowering students with valuable skills in biomedical electronics and cardiovascular diagnostics.

## Experimental Methods

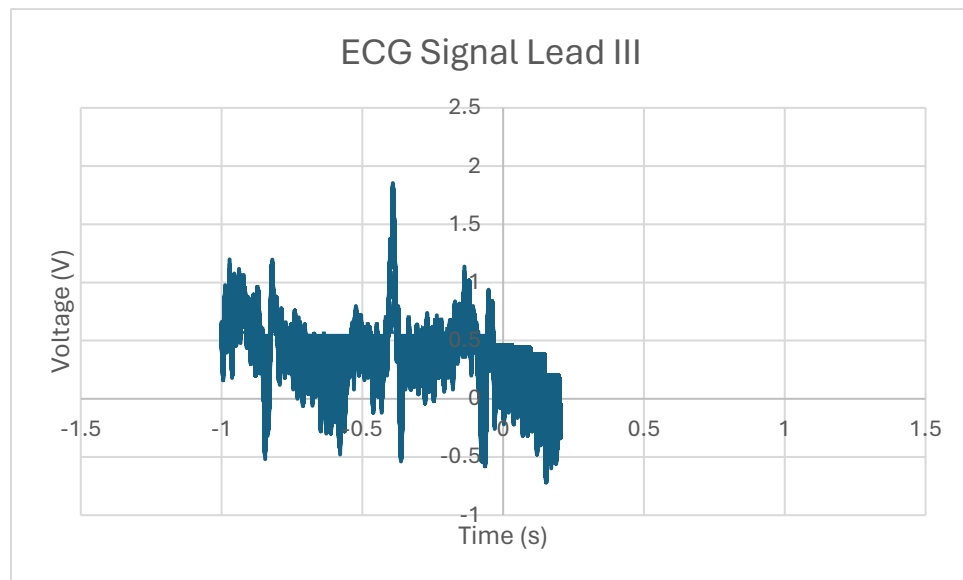
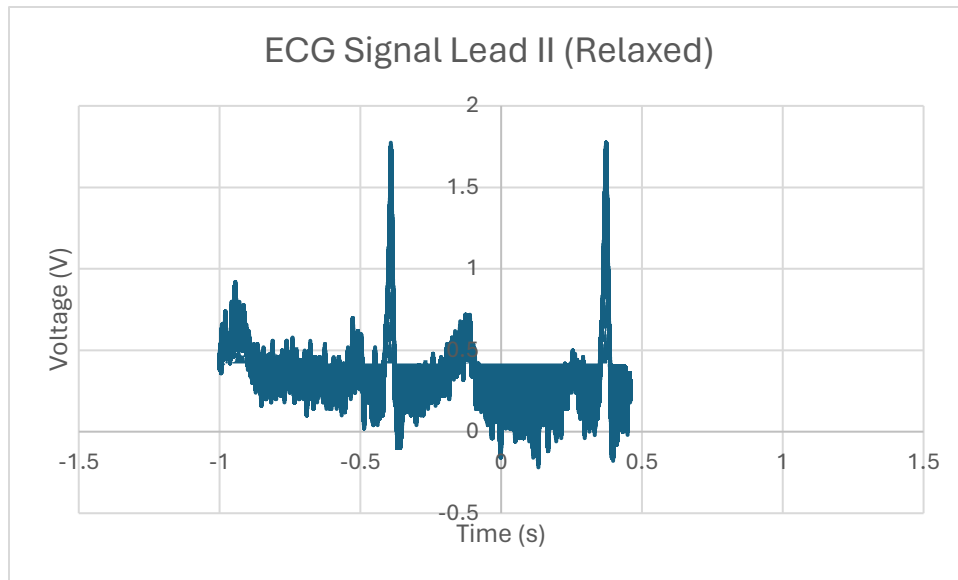
In our laboratory experiment, we began by verifying the functionality of the ECG amplifier circuit using a frequency generator to ensure signal amplification. Once confirmed, electrodes were placed on the subject's wrists and ankle following the Lead I configuration (figure 3), where the positive electrode was positioned on the left hand, the negative electrode on the right hand, and the ground electrode on the ankle. These electrodes were integrated with 10k  $\Omega$  resistors in series and connected to the ECG amplifier. Subsequently, the amplified signals were recorded using an oscilloscope under two conditions: when the subject was at rest and when the subject clenched its fists. This procedure was repeated with electrodes configured according to Lead II, with the positive electrode positioned on the left leg. Through these steps, we aimed to capture variations in bio-potentials and analyze the resulting ECG signals to gain insights into cardiac activity and physiological responses.

## Results and Discussion

- Plot ECG signal from lead I and lead II from the case when you are relaxed.
- Calculate and plot ECG signal for lead III from measured value of lead I and II For ECG data.







- From lead I:
  - Measure the duration of each portion of the waves.
  - Measure the amplitude of each wave.
  - Measure the beating frequency of your heart.

|          | Amplitude | Duration (s) |
|----------|-----------|--------------|
| P Wave   | 0.35 V    | 0.31 s       |
| QRS Wave | 0.91 V    | 0.088 s      |
| T Wave   | 0.41 V    | 0.144 s      |

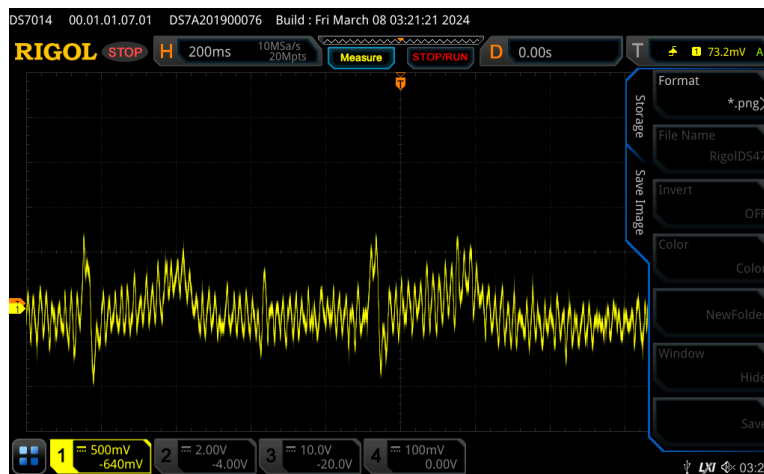
Looking at the previous table and adding the duration for each wave, we calculate the period of the heartbeat to be 0.542 seconds. To get the beats per minute (bpm), we take the reciprocal and times it by 60. So, the bpm is the following:

$$\frac{1}{\text{period (s)}} * 60 \frac{\text{s}}{\text{min}} = 110 \text{ bpm}$$

- Plot the ECG signals from lead I for the case when you hold your hands tight and when you are relaxed. Discuss the difference in the signal. Which signal has more “noise,” where does the “noise” come from?



4: ECG Signal Lead 1 (Relaxed) from Rigol Oscilloscope



5: ECG Signal Lead 1 (Clenched) from Rigol Oscilloscope

There are noticeably more noises in the clenched fist graph. This is due to the increased action potentials firing in the muscle of the hand when you are forcing yourself to clench your fist.

- Find the mean electrical field orientation of QRS complex.

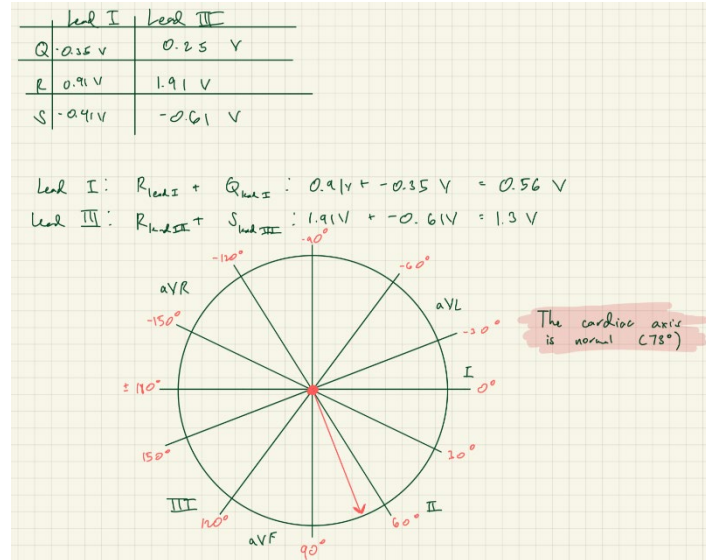


Figure 6: Mean Electrical Axis

- The mean QRS vector roughly corresponds to the anatomical longitudinal axis of the heart. Is the orientation you found reasonable? Why don't you need to use the Lead III data?

Plugging in the values of Lead I and Lead II to find the mean electrical axis of the QRS complex we find the value to be 73 degrees. This calculation was done on an online heart axis calculator since I am not able to make the graph to scale. The found value of 73 degrees is within the normal range of -30 to 90 degrees, so yes, the value is reasonable. To find the orientation it is not necessary to use Lead III, it can be found with any combination of 2 leads, it just happened that I used Lead I and Lead III in this case but using Lead I and Lead II would have been fine as well.

## Conclusion

This laboratory experiment provided insights into the interplay between Leads I, II, and III in the context of ECG signal acquisition and analysis. We gained proficiency in interpreting bio-electric signals and understanding their significance in clinical settings. By leveraging data from Leads I and II, we successfully derived accurate Lead III information, demonstrating the efficacy of signal processing techniques in refining ECG waveforms. Finally, we learn how to use the QRS complex to monitor said cardiac activity.